ORGAN SYSTEMS 3
HUMAN GROSS ANATOMY AND EMBRYOLOGY
LEARNING OBJECTIVES

The most difficult aspect of studying anatomy is organizing your time to allow uptake and retention of a large quantity of information in a limited amount of time. To help you organize yourself, the following is a list of learning (study) objectives. The general objectives, listed first, apply to all course material. The specific structure objectives are important guides to pertinent topics in each of the integrated courses. Consider these carefully as you progress through the relevant dissections and lectures. Remember, all examination material will be derived from the study objectives.

GENERAL COURSE OBJECTIVES

1. Spell and pronounce anatomical terminology correctly and precisely.
2. Identify, differentiate and/or delimit the body regions, organ systems, organs, and subdivisions of the organs of the human body.
3. Describe the *in situ* relationships of the major structures in the human body.
4. Delineate the surface projections of clinically significant internal structures and their components.
5. Correlate structure and function of the major structures in the human body.
6. Integrate the gross and microscopic continuities of organs and organ systems within and among body regions.
7. Describe how organogenesis of systems accounts for the formation and disposition of organs in the adult body.
8. Describe the features of and developmental bases for the major congenital malformations of organs and regions.
9. Identify the major structures of the human body in standard medical images.
10. Predict the clinical significance and functional consequences of the major pathological conditions of normal human anatomy.
11. Identify the major structures of the human body in dissection.
12. Develop a three-dimensional working knowledge of the composition of the human body by the integration of its component parts.

SPECIFIC STRUCTURE OBJECTIVES - A SAMPLE PLAN

The following outline provides a precise method for organizing anatomical information. You may find this an effective study methodology. For each structure encountered in class, ask yourself the appropriate questions listed. Remember to use the outline only in the context of structures presented in dissection and lecture. Do not attempt to assimilate every structure noted in your textbook.

<u>BONE</u>

1. Location.
2. Articulation(s).
3. Developmental type (membrane or endochondral).
4. Growth centers (primary and secondary).
5. Maturation age.
6. Muscle, tendon, ligament attachments.
7. Neurovascular or other structures housed or conveyed or having close topographic relationship.
8. Surface projection and palpable points.
9. Effect of specific fractures.

<u>JOINT</u>

1. Type (classification and degrees of freedom permitted).
2. Bones involved (including specific parts of bones concerned).
3. Extent and attachment of capsule and lining.
4. Supporting extra-articular ligaments and muscles.
5. Intra-articular structures (articular disc and/or ligaments).
6. Movement caused by each muscle affecting joint.
7. Innervation.
8. Blood supply and drainage.

<u>MUSCLE</u>

1. Origin
2. Insertion.
3. Structure (subdivisions, fiber direction, pinnation, etc.).
4. Action (on which joint or joints - type of action).
5. Innervation.
6. Blood supply.
7. Relations.
8. Antagonist(s).
9. Effect of loss.
10. Surface projection and/or palpation.

BLOOD VESSEL

1. Artery or vein.
2. Origin (from what other vessel).
3. Course and relations (subdivide into parts if needed).
4. Named branches (where each originates).
5. Area supplied or drained by each named branch (give specific area).
6. If any branches have named divisions, list these.
7. Companion structures.
8. Major anastomoses and collateral routes.
9. Effect of blockage.
10. Surface projection and/or palpation.

NERVE

1. Functional components
2. Origin of fibers.
3. Course and relations of fibers.
4. Target(s) of fibers.
5. Location of cell bodies of fibers.
6. If autonomic (visceral) innervation:
   a. Parasympathetic vs. Sympathetic.
   b. Origin and course of preganglionic fibers.
   c. Location of synapses.
   d. Origin, course, and target of postganglionic fibers.
   e. Afferent companions and projections.
7. Effect of loss.
8. Surface projection and/or palpation.

SPACE OR AREA

1. Location.
2. Boundaries (structures which make up boundaries).
3. Contents and/or transients.
4. Surface projection and/or palpation.

ORGAN

1. Definition (type).
2. Location.
3. Relations (contacts).
4. Structure (size, shape, lobes, fissures, ducts, stroma, parenchyma, etc.).
5. Function.
6. Blood supply and drainage.
7. Innervation.
8. Lymphatic drainage.
9. Developmental history.
10. Effect of loss or pathology.
11. Surface projection and/or palpation.

LYMPHATIC DRAINAGE

1. Area(s) drained.
2. Course of channels from area drained to nodal groups.
3. Names of node groups.
4. Location of node groups.
5. Routes of drainage from node groups to next major node groups.
6. Drainage point with venous system.
7. Palpation points of nodal groups.
8. Function.

HEAD AND NECK

OSTEOLOGY AND ARTHROLOGY

1. Identify the bones of the skull, hyoid, and cervical spine, and their major features, in dry osteology specimens and in standard medical imaging. Describe the functional aspects of these structures.

DEVELOPMENT OF THE BRANCHIAL APPARATUS AND SKULL

1. Define the terms "branchial (pharyngeal) apparatus/basket," "branchial arch," "branchial pouch," and "branchial cleft/groove." Describe the development of the branchial apparatus, including the role of neural crest, and list the components of the pharyngeal basket.

2. Trace the developmental fate of each component of the branchial apparatus. Describe the features and embryological bases of the major congenital defects in the branchial apparatus.

3. Trace the development of the tongue, noting its embryonic links to the branchial apparatus. Describe the features and embryological basis of the major congenital defects in the tongue.

4. Trace the development of the thyroid gland, noting the embryonic and adult links between it and the tongue. Describe the features and embryological basis of the major congenital anomalies in the thyroid gland.

5. Outline the features of the developing skull, giving definitions of neurocranium, chondrocranium, and viscerocranium. Identify the major fontanelles and explain their functional significance. Define the features and developmental bases of the major congenital defects in the skull.

ORGANIZATION OF THE FACE AND SCALP

1. Describe the boundaries of the face and scalp.

2. Describe the development of the face and palate. Identify the features and developmental bases of the major craniofacial malformations, including clefts and holoprosencephaly.

3. Describe the structure of the scalp.

4. Identify the major muscles of facial expression and their actions.

5. Describe the innervation of the face and scalp. Predict the deficit expected following injury to each of the major nerve branches.

6. Trace the flow of blood into the face and scalp.

7. Trace the pattern of venous drainage of the face and scalp.

8. Describe the morphology and general relationships of the parotid gland. Describe the clinical significance of the relationship between the parotid gland, its duct, and the extracranial distribution of the facial nerve.

INTERIOR OF THE SKULL

1. Identify the cranial fossae, the major bony components and boundaries of each, and the major contents of each.

2. Identify the meninges surrounding the brain and the folds of dura mater which subdivide the cranium. Explain the functional/clinical significance of this arrangement. Describe the innervation of the dura mater.

3. Trace the flow of blood into the cranial cavity, indicating the major anastomoses and collateral routes. Describe the formation of the cerebral arterial circle, and explain its functional and clinical significance.

4. Identify the dural venous sinuses, indicating their relations to the cranial meninges. Trace the communications of the venous sinuses with the extracranial venous system, and explain the functional/clinical significance of this arrangement.

5. Explain the anatomical basis for epidural, subdural, and subarachnoid cranial hemorrhages.

6. Identify each of the cranial nerves as it leaves the cranial cavity. Indicate the relationships of each nerve to particular cranial fossae, to the folds of dura mater, and to venous sinuses along its course. Also indicate the exact pathway taken in exiting the cranium (e.g., foramina traversed), and any major accompanying structures (e.g., blood vessels).

ORBIT AND EYE

1. Identify the bony elements of the orbit. Indicate the major extraorbital structures lying superior, inferior, medial, and lateral to the orbit, and the position of the eyeball relative to the orbit.

2. Identify the extraocular muscles, their actions, and their nerve supply. Predict the functional deficit resulting from damage to each muscle. Describe the method of clinical testing of the individual extraocular muscles and their nerves.

3. Identify the major nerves of the orbit, their functional components, their main branches, and their destinations. Predict the functional deficit resulting from damage to each nerve.

4. Trace the flow of blood into and out of the orbit and orbital structures.

5. Identify the main components of the eyelids. Describe the mechanics of movement of the eyelids.

6. Identify the components of the lacrimal apparatus. Trace the pathway a tear takes from the lacrimal gland to the inferior nasal meatus.

7. Follow the course of autonomic nervous supply to orbital structures, indicating the pre- and postganglionic sources of innervation. Differentiate sympathetic and parasympathetic functions.

ORGANIZATION OF THE NECK

1. Describe the arrangement and functions of the deep cervical fascia with respect to the organization of the neck.

2. Identify the major triangles of the neck, the boundaries of each, and the major contents of each.

3. Identify the muscles in the neck, including their attachments, innervation, and major actions.

4. Identify the carotid sheath and its contents. Describe the topographic relations of the sheath as a whole and of each constituent.

5. Identify the thyroid and parathyroid glands, their vascular and nervous supply, and their relations to each other and surrounding cervical structures.

6. Trace the course of cranial nerves IX, X, XI, and XII in the neck. Describe the relationship of each nerve to its major neighboring structures. Identify the functional components of each nerve. Describe the deficit expected from lesion of each.

7. Describe the formation of the cervical and brachial plexuses of nerves, noting the spinal segments of origin, relations to surrounding cervical structures, and distribution of peripheral branches. Note the formation, relations, and distributions of the phrenic nerve and the ansa cervicalis.

8. Describe the location and relations of the cervical sympathetic trunk and its ganglia.

9. Trace the flow of blood through the subclavian artery and its major branches. Note the regions supplied by each branch, the relationship of the branches to surrounding structures, and anastomoses between the branches.

10. Trace the flow of blood through the carotid arterial tract. Note the regions supplied by each of the major branches, the relationship of the branches to surrounding structures, and anastomoses between branches. Describe the locations and functions of the carotid sinus and carotid body.

11. Trace the flow of blood through the jugular system of veins, noting the regions drained by each of the tributaries, and interconnections between the major veins.

<u>TEMPORAL, INFRATEMPORAL, PTERYGOPALATINE REGIONS</u>

1. Define the temporal, infratemporal, and pterygopalatine fossae. Identify the major structures contained in each region.

2. Identify the muscles of mastication, their attachments, sources of innervation, and major actions in chewing.

3. Trace the flow of blood through the maxillary artery and its major branches. Identify the regions supplied and the anastomoses between branches.

4. Identify, and trace the courses of, the nerves that traverse the above fossae. Identify the functional components of each nerve, their sources, and their areas of termination.

NEURAL ORGANIZATION

1. Identify the 12 pairs of cranial nerves. Define the functional components of the cranial nerves.

2. Identify the functional component(s) contained within and/or conveyed by each of the cranial nerves.

3. Trace the course of each of the cranial nerves from its origin at the base of the brain to its final destination(s). Indicate the avenue(s) taken in exiting the skull, and the peripheral relations of each nerve. Predict the functional deficit(s) expected from destruction of each nerve.

4. Identify the sources of autonomic innervation to the head.

5. Trace the pathways of preganglionic and postganglionic autonomic neurons in the head from their origins to their final destinations. Indicate specific synapse points.

6. Describe the primary functions governed by each autonomic component in the head.

LYMPHATIC DRAINAGE

1. Trace the routes of lymphatic drainage in the head and neck. Indicate the major aggregations of lymph nodes and their relations to neighboring structures.

ORAL CAVITY AND PHARYNX

1. Define the boundaries and subdivisions of the oral cavity and pharynx. Describe the bony, muscular, neurovascular, and glandular relations of each.

2. Identify the muscles of the oral floor and walls, soft palate and auditory tube, and pharynx, and their attachments. Identify the innervation of these muscles.

3. Identify the extrinsic and intrinsic muscles of the tongue. Describe the sensory and motor nerve supply to the tongue. Predict the deficit expected to follow an injury to each nerve.

4. Describe the mechanism of swallowing. Note the sequence of events, the muscles responsible for each event, and the nerves controlling each event.

5. Describe the pattern of innervation of the oropharyngeal region. Identify the source, region supplied, and the functional components of each nerve.

6. Trace the arterial supply to the oropharyngeal region. Identify the major arteries, their territories, and any significant anastomoses.

7. Describe the location, innervation, lymphatic drainage, secretory drainage, and general relationships of the submandibular and sublingual salivary glands.

8. Describe the location, lymphatic drainage, and general relationships of the oropharyngeal tonsils.

NASAL REGION

1. Describe the basic structure and relationships of the external nose and nasal cavities.

2. Identify the paranasal sinuses, noting the drainage route of each into the nasal cavity. Describe the relationship of each sinus to the surrounding oral, orbital, and cranial cavities.

3. Describe the pattern of innervation of the nasal region and paranasal sinuses.

4. Trace the arterial supply to the nasal region.

LARYNX

1. Identify the major morphological features of the larynx.

2. Identify the major intrinsic muscles of the larynx. Determine the actions of these muscles and the roles of these actions in sound production.

3. Trace the courses of the neurovascular supply of the larynx. Predict the functional consequences of damage to the different nerves comprising this innervation.

4. Describe the major topographic relationships of the larynx and its neurovascular supply. Note the clinical significance of such arrangements.

EXTERNAL AND MIDDLE EAR

1. Briefly describe the development of the external and middle ear regions.

2. Identify the boundaries and components of the external ear. Describe the innervation of the region.

3. Identify the boundaries and contents of the middle ear. Indicate the relationships of the major neighboring structures.

4. Identify the muscles of the middle ear, their actions, and their sources of innervation.